Yue Ma

LinkedIn • GitHub • Google Scholar • Website • yuema9810@gmail.com

RESEARCH PROFILE

Postdoctoral researcher building resource-efficient agents for science and engineering. My work spans autonomous research systems, agent evaluation, statistical modeling, and scientific ML, measuring outcomes, decision trajectories, uncertainty, and resource use.

EDUCATION & ACADEMIC POSITIONS

Postdoctoral Researcher, Halıcıoğlu Data Science Institute, UC San Diego — Mar 2026 - Present

Ph.D. in Particle Physics, University of California, San Diego — Oct 2020 - Feb 2026

B.S. in Physics, Kuang Yaming Honors School, Nanjing University — Aug 2016 - Jun 2020

AGENT SYSTEMS & SCIENTIFIC EVALUATION

SIDERIUS: Resource-Aware Autonomous Research

Feb 2026 - Present

Sole architect and developer. Built long-horizon scientific ML infrastructure; evaluated on TIDMAD and reused for TESS, LIGO, and Project 8 physics time-series tasks.

- **Composable architecture:** Designed seven capabilities with typed, caller-independent contracts for fixed workflows, LLM orchestration, and direct human intervention.
- **Scientific validity:** Separated optimization, monitoring, ranking, and supporting evidence. Combined online scoreability/health checks with offline LLM behavioral review to detect collapse and gaming, using a frozen rubric without feeding judgments back into search.
- **Resource-aware exploration:** Built GPU cost probes, runtime/VRAM budgets, feasibility checks, monitoring, and bounded retries; measured costs and failures inform subsequent proposals.
- **TIDMAD dark-matter denoising benchmark:** Best orchestrated configuration (without Data Analysis) achieved **63.8% valid proposals vs. 1.6%** for Codex with the same GPT backend, and **over two orders of magnitude higher aggregate signal-to-noise ratio** for best valid models under the benchmark protocol.

SciTra: Scientific-Agent Evaluation & Trajectories

Jul 2026 - Present

Founder and Coordinator. Built a collaboration of approximately 100 researchers across 20+ institutions, with 50+ evaluation tasks in development; coordinate domain, evaluation, and infrastructure teams.

- **Evaluation methodology:** Lead research on outcomes, resource use, and decision trajectories. Represent scientific actions and decisions using domain knowledge and human-agent interactions to support comparisons across agents, tasks, and domains.
- **Evidence infrastructure:** Built cross-vendor trajectory instrumentation and coordinate requirements for a Rust-based runtime. Online records capture interactions, tool use, Git changes, costs, outputs, and failures; task-specific evaluation processes the evidence offline.

TECHNICAL SKILLS

Programming & ML	Python, PyTorch, NumPy, Pandas, SciPy, scikit-learn, C/C++, Numba, Bash, Linux, Git
Agents & Statistics	Multi-agent orchestration, benchmark design, trajectory analysis, Bayesian and likelihood inference, uncertainty propagation, hypothesis testing
Infrastructure	Pydantic, FastAPI, pytest, GitHub Actions, SLURM/HPC, ETL pipelines, resource monitoring and performance optimization

SELECTED TALKS & SERVICE

Invited SIDERIUS talk and scientific-agent Birds-of-a-Feather session, Fast ML for Science, UC San Diego (Sep 2026). Session chair, APS Global Physics Summit (Mar 2026).

STATISTICAL MODELING, EVALUATION & SCIENTIFIC COMPUTING

Measurement & Inference in XENONnT

Oct 2020 - Feb 2026

— Developed Bayesian and maximum-likelihood inference frameworks and Monte Carlo workflows for uncertainty propagation, sensitivity estimation, and hypothesis testing in rare-event searches.

— **Corresponding author** on the first solar-neutrino measurement through keV-scale neutrino-electron scattering; contributed statistical inference, data processing, and detector response modeling. Core contributor to dark-matter and electronic-recoil searches through event selection, efficiency/background modeling, and detector calibration.

Data Infrastructure: strax & straxen

Jan 2024 - Feb 2026

Key maintainer of strax and straxen, processing petabyte-scale detector data for 200+ collaborators. Reduced ETL latency by 40% through vectorized NumPy/Pandas workflows, Numba acceleration, and memory-efficient processing.

SCIENTIFIC MACHINE LEARNING

— **Density-guided conditional neural processes:** First-author work on detector efficiency estimation, connecting probabilistic ML with detector-response modeling (2026; publication below).

— **Adaptive neural processes for extreme forecasting** (Nov 2024 - Jun 2025): Conceived and derived AdaCNP for few-shot adaptation to unseen extreme regimes without retraining; achieved 22% lower MSE and best negative log-likelihood among Neural Process baselines on two power-system datasets.

— **Multi-fidelity surrogates for simulation** (Jul - Oct 2024): Combined Gaussian and Neural Processes for rare-event simulation; reduced simulation cost by 96.7% and improved background-suppression efficiency by 66.5%. Core algorithms and derivations for RESuM, ICLR 2025 Spotlight (**acknowledged contribution; not a coauthor**).

EVALUATION RESOURCES & HUMAN-AGENT TOOLS

— **Leaderboard of Benchmarks:** Compare evaluation instruments by capability coverage, frontier evolution, saturation, and cost. **SciEval:** Bilingual map of scientific-agent benchmarks, measurement scope, and improvement opportunities.

— **PairTeX:** In-place manuscript review and structured agent feedback for LaTeX repositories. **Structured Coding:** Planning, implementation, validation, and review workflows for coding agents.

SELECTED PUBLICATIONS

— **Yue Ma**, Lucas Venetoulias, Akbota Assan, Aobo Li (2026). Beyond a Better Score: Long-Horizon Agentic ML Development and Evaluation Protocol for Physics Time Series. Submitted to ICLR; arXiv:2610.14024.

— **Yue Ma**, Aobo Li (2026). Density-Guided Conditional Neural Processes for Detector Efficiency Estimation. ML4PS Workshop 2026; arXiv:2609.13593.

— Zijun Liu, **Yue Ma**, Christopher A. Theissen (2026). What does self-supervision add to engineered features for stellar light curves? ML4PS Workshop 2026; arXiv:2610.14283.

— E. Aprile et al. (XENON Collaboration, including **Yue Ma**) (2026). First Measurement of Solar Neutrinos through Elastic Neutrino-Electron Scattering at the keV Scale. Preprint, arXiv:2608.29450. **Corresponding author.**

— Yiming Huang et al. (including **Yue Ma**) (2026). Can LLMs Discover Scientific Laws in Real and Parallel Worlds? Submitted to ICLR; arXiv:2609.01552.

— Hejia Geng, Zesen Huang, et al. (including **Yue Ma**) (2026). ScienceIDE: Turning World's Scientific Codebase into Agent Learnable Environments. Submitted to ICLR; arXiv:2609.19134.

— Chenxi Hu, **Yue Ma**, Yifan Wu, Yunhe Hou (2026). Resilient Load Forecasting under Climate Change: Adaptive Conditional Neural Processes for Few-Shot Extreme Load Forecasting. Preprint, arXiv:2602.04609.

— E. Aprile et al. (XENON Collaboration, including **Yue Ma**) (2023). First Dark Matter Search with Nuclear Recoils from the XENONnT Experiment. Phys. Rev. Lett. **131**, 041003.

— E. Aprile et al. (XENON Collaboration, including **Yue Ma**) (2022). Search for New Physics in Electronic Recoil Data from XENONnT. Phys. Rev. Lett. **129**, 161805.